

ALPHA · α

IOMINDS.AI · DETAILED CURRICULUM

Alpha — Junior IoT & AI Engineers

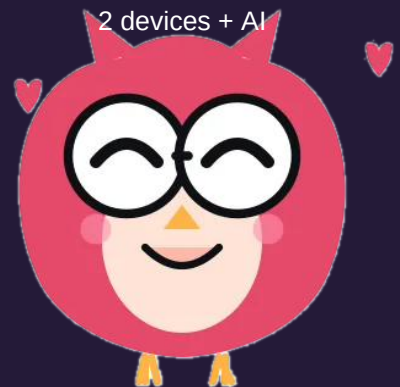
Basic level · Ages 6–9 · 10 live sessions · Smart Night Light → Automated Plant Doctor → first AI model

Level
Basic

Ages
6–9

Sessions
10 live

Builds
2 devices + AI



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Course overview

Alpha is the flagship **Junior IoT & AI Engineers** journey — a 10-session, project-based program in which every learner physically assembles working devices and walks away with functioning projects they built and understand. The sequence is intentional: students progress from making a single LED respond to light, to automating real-world care of a living plant, to recognising patterns with AI.

Level	Alpha (α) — Basic
Ages	6–9 years (Grades 4–5 friendly)
Format	10 live, small-batch sessions — learn-by-doing, no lectures
Hardware	Professional-grade IoT kit (ESP32 / Arduino) delivered to the doorstep before class 1
Languages	Simplified C++ (Arduino) with visual tools: Tinkercad, Google Teachable Machine
Outcome	Two working devices carried home + a first trained AI model + Grand Demo showcase

Part 1 — The Smart Night Light (Classes 1–4)

Learning goal: how "things" sense the environment and react to it.

Class 1	Hello, World!	Why the internet matters in daily life; what IoT is (giving everyday objects a smart voice); components of IoT — sensors, connectivity, data processing, user interface; benefits vs. threats (privacy & security). Hands-on: connecting the controller and blinking the first LED.
Class 2	Sensing the Dark	Hardware fundamentals: what a computer is, microcontrollers as tiny one-job brains, development boards (Arduino Uno, ESP32, micro:bit); sensors as the five senses of machines. Hands-on: wiring the LDR light sensor and reading live brightness values as students cover it with a hand.
Class 3	Logic & Decisions	The if-then rule — the heart of every smart device. Digital vs. analog signals, input vs. output, basic commands (pinMode, digitalWrite, analogRead, delay) and the setup/loop structure. Hands-on: programming the LED to turn on only when the room is dark.
Class 4	The Smart Lamp Build	Full assembly: LDR + ESP32 + LED in a cardboard or 3D-printed housing. Calibrating the darkness threshold, testing auto-on behaviour, and troubleshooting circuits. Outcome: a functional night light for the student's own bedroom.

Part 2 — The Automated Plant Doctor (Classes 5–8)

Learning goal: data, sustainability, and automated action (smart agriculture).

Class 5	Thirsty Plants	Introducing the soil moisture sensor; how machines "feel" dryness. Experiment: comparing readings from dry soil vs. a wet sponge, and charting the values.
Class 6	The Water Pump & Relays	Actuators — the muscles of machines: motors, pumps, buzzers, LEDs. How relays let small computers safely control "big" electricity. Hands-on: clicking a relay to switch the 5V mini water pump.
Class 7	Integration (The Brain)	Bringing sense + think + act together. Calibration and the rule: if soil moisture < 30%, turn on the pump for 5 seconds. Debugging real electrical and logical errors.
Class 8	The Plant Doctor Assembly	Setting up tubing, sensor and pump on a real pot. The finished device waters the plant automatically while the student sleeps. Discussion: how the same idea scales to smart farms.

Part 3 — Introduction to AI & Showcase (Classes 9–10)

Learning goal: how AI elevates "smart" things into "intelligent" ones.

Class 9	AI Training Day	AI as pattern recognition. Using Google Teachable Machine, students train a model to distinguish a healthy leaf from a wilted one with their own webcam — then connect the idea back to the Plant Doctor.
Class 10	The Grand Demo	Students present their working night light and plant doctor to parents and peers; "future talk" connects classroom sensors to large-scale projects in smart cities, healthcare and agriculture. Certificates awarded.

The hardware kit (per student)

1×	ESP32 Microcontroller	The brain — Wi-Fi/Bluetooth enabled (Arduino Uno variant available). Reused across the full project library.
1×	Light Sensor (LDR)	The eyes of the Smart Night Light.
1×	Soil Moisture Sensor	The fingertips of the Plant Doctor.
1×	Bright / RGB LED pack	Primary actuator for visual output.
1×	5V Mini Water Pump + tubing	Submersible pump for automated plant watering.
1×	Relay Module	Safely switches the pump from microcontroller signals.

set	Breadboard, jumper wires (M-F), USB cable	Wiring and prototyping accessories — no soldering required, ever.

Every enrolment includes

Hardware kit included with the course · lifetime access to study material · lifetime support for queries · lifetime updates on national and international competitions · course completion certificate.

Book a free trial class at iominds.ai — hello@iominds.ai